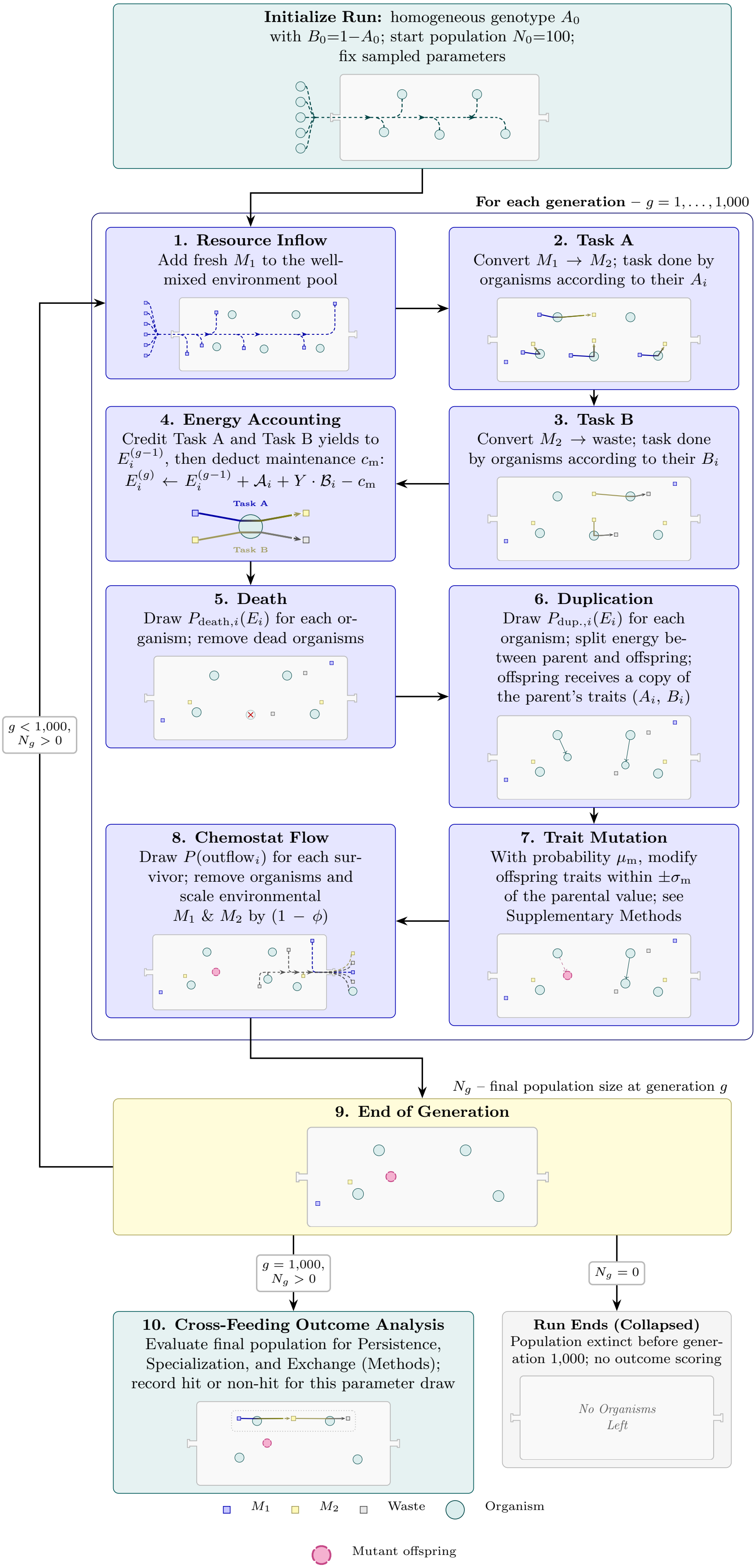
Initialize Run: homogeneous genotype A_0 with $B_0=1-A_0$; start population $N_0=100$; fix sampled parameters
For each generation – $g = 1, \ldots, 1{,}000$
1. Resource Inflow
Add fresh M_1 to the well-mixed environment pool
2. Task A
Convert $M_1 \rightarrow M_2$; task done by organisms according to their A_i
3. Task B
Convert $M_2 \rightarrow$ waste; task done by organisms according to their B_i
4. Energy Accounting
Credit Task A and Task B yields to $E_i^{(g-1)}$, then deduct maintenance c_m:
$E_i^{(g)} \leftarrow E_i^{(g-1)} + \mathcal{A}_i + Y \cdot \mathcal{B}_i - c_m$
Task A
Task B
5. Death
Draw $P_{\text{death},i}(E_i)$ for each organism; remove dead organisms
6. Duplication
Draw $P_{\text{dup.},i}(E_i)$ for each organism; split energy between parent and offspring; offspring receives a copy of the parent's traits (A_i, B_i)
7. Trait Mutation
With probability μ_m, modify offspring traits within $\pm\sigma_m$ of the parental value; see Supplementary Methods
8. Chemostat Flow
Draw $P(\text{outflow}_i)$ for each survivor; remove organisms and scale environmental M_1 & M_2 by $(1 - \phi)$
$g < 1{,}000$, $N_g > 0$
N_g – final population size at generation g
9. End of Generation
$g = 1{,}000$, $N_g > 0$
$N_g = 0$
10. Cross-Feeding Outcome Analysis
Evaluate final population for Persistence, Specialization, and Exchange (Methods); record hit or non-hit for this parameter draw
Run Ends (Collapsed)
Population extinct before generation 1,000; no outcome scoring
No Organisms Left
M_1
M_2
Waste
Organism
Mutant offspring